

# Replay-Aware CVRP Benchmark Report

## Overview

- Condition count: 4.
- Best final transfer gap: cvrp\_no\_replay.
- Best CVRPLIB holdout gap: cvrp\_failure\_replay.
- Best synthetic holdout gap: cvrp\_no\_replay.

## Run Metadata

- run\_name: run\_20260513\_193947\_m
- started\_at\_local: 2026-05-13 19:39:47
- finished\_at\_local: 2026-05-13 19:42:50
- duration\_hhmm: 00:03
- duration\_seconds: 182.974
- seed\_offset: 12000
- replicate\_label: m
- judge\_status: enabled
- judge\_provider: openai
- judge\_model: gpt-4.1-mini

## Condition Comparison

| Condition                       | Replay  | Selection    | Compression | Final CVRPLIB Gap | Final Synthetic Gap | Final Transfer Gap | Mean Accepted Novelty | Mean Complexity | Adaptation Efficiency |
|---------------------------------|---------|--------------|-------------|-------------------|---------------------|--------------------|-----------------------|-----------------|-----------------------|
| cvrp_no_replay                  | none    | score_only   | False       | 0.218068          | 0.005376            | 0.123538           | 0.767949              | 0.78            | 0.021668              |
| cvrp_random_replay              | random  | score_only   | False       | 0.24523           | 0.005376            | 0.138628           | 0.890782              | 0.58            | 0.0061                |
| cvrp_failure_replay             | failure | score_only   | False       | 0.215701          | 0.01914             | 0.128341           | 0.826087              | 0.78            | 0.057228              |
| cvrp_failure_replay_compression | failure | novelty_gate | True        | 0.243813          | 0.005376            | 0.137841           | 0.0                   | 0.58            | 0.0                   |

## Condition Notes

### cvrp\_no\_replay

- Replay mode: none with selection mode score\_only.
- Final transfer gaps: CVRPLIB 0.218068, synthetic 0.005376, combined 0.123538.

- Accepted-epoch count 2, mean accepted code novelty 0.767949, and final complexity 0.78.
- Adaptation efficiency 0.021668 and archive sizes {'worst\_cases': 5, 'failure\_cases': 5, 'adversarial\_layouts': 4} / elite 2.
- Panel heldout\_cvrplib mean gap 0.218068 across 5 instances; family means: A=0.095986, B=0.237997, E=0.247601, P=0.270758.
- Panel synthetic\_holdout mean gap 0.005376 across 4 instances; family means: alternating\_belt=0.021505, clustered\_demand=0.0, corridor\_split=0.0, radial\_heavy=0.0.
- Worst recent training cases: [{"best\_known\_cost": 458, "cost": 611, "family": "P", "name": "P-n40-k5", "optimality\_gap": 0.334061}, {"best\_known\_cost": 937, "cost": 1236, "family": "A", "name": "A-n44-k6", "optimality\_gap": 0.319104}, {"best\_known\_cost": 672, "cost": 807, "family": "B", "name": "B-n31-k5", "optimality\_gap": 0.200893}]

### **cvrp\_random\_replay**

- Replay mode: random with selection mode score\_only.
- Final transfer gaps: CVRPLIB 0.24523, synthetic 0.005376, combined 0.138628.
- Accepted-epoch count 2, mean accepted code novelty 0.890782, and final complexity 0.58.
- Adaptation efficiency 0.0061 and archive sizes {'worst\_cases': 5, 'failure\_cases': 5, 'adversarial\_layouts': 4} / elite 1.
- Panel heldout\_cvrplib mean gap 0.24523 across 5 instances; family means: A=0.121291, B=0.254873, E=0.293666, P=0.301444.
- Panel synthetic\_holdout mean gap 0.005376 across 4 instances; family means: alternating\_belt=0.021505, clustered\_demand=0.0, corridor\_split=0.0, radial\_heavy=0.0.
- Worst recent training cases: [{"best\_known\_cost": 458, "cost": 587, "family": "P", "name": "P-n40-k5", "optimality\_gap": 0.281659}, {"best\_known\_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality\_gap": 0.135539}, {"best\_known\_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality\_gap": 0.096939}]

### **cvrp\_failure\_replay**

- Replay mode: failure with selection mode score\_only.
- Final transfer gaps: CVRPLIB 0.215701, synthetic 0.01914, combined 0.128341.
- Accepted-epoch count 2, mean accepted code novelty 0.826087, and final complexity 0.78.
- Adaptation efficiency 0.057228 and archive sizes {'worst\_cases': 5, 'failure\_cases': 5, 'adversarial\_layouts': 4} / elite 2.
- Panel heldout\_cvrplib mean gap 0.215701 across 5 instances; family means: A=0.094241, B=0.187483, E=0.259117, P=0.350181.
- Panel synthetic\_holdout mean gap 0.01914 across 4 instances; family means: alternating\_belt=0.021505, clustered\_demand=0.0, corridor\_split=0.0, radial\_heavy=0.055054.
- Worst recent training cases: [{"best\_known\_cost": 784, "cost": 1087, "family": "A", "name": "A-n32-k5", "optimality\_gap": 0.38648}, {"best\_known\_cost": 458, "cost": 600, "family": "P", "name": "P-n40-k5", "optimality\_gap": 0.310044}, {"best\_known\_cost": 672, "cost": 829, "family": "B", "name": "B-n31-k5", "optimality\_gap": 0.233631}]

### **cvrp\_failure\_replay\_compression**

- Replay mode: failure with selection mode novelty\_gate.
- Final transfer gaps: CVRPLIB 0.243813, synthetic 0.005376, combined 0.137841.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst\_cases': 5, 'failure\_cases': 5, 'adversarial\_layouts': 4} / elite 1.
- Panel heldout\_cvrplib mean gap 0.243813 across 5 instances; family means: A=0.11082, B=0.269043, E=0.268714, P=0.301444.
- Panel synthetic\_holdout mean gap 0.005376 across 4 instances; family means: alternating\_belt=0.021505, clustered\_demand=0.0, corridor\_split=0.0, radial\_heavy=0.0.
- Worst recent training cases: [{"best\_known\_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality\_gap": 0.318777}, {"best\_known\_cost": 937, "cost": 1073, "family": "A", "name": "A-n44-k6", "optimality\_gap": 0.145144}, {"best\_known\_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality\_gap": 0.096939}]

## Judge Appendix

### Summary of CVRP Benchmark Conditions

| Condition                           | Replay Mode                 | Mean Held-out CVRPLIB Gap  | Mean Synthetic Holdout Gap | Mean Transfer Gap | Code Novelty (mean) | Adaptation Efficiency | Replay Compression | Final Complexity | Notes on Replay Type           |
|-------------------------------------|-----------------------------|----------------------------|----------------------------|-------------------|---------------------|-----------------------|--------------------|------------------|--------------------------------|
| **cvrp_no_replay**                  | none                        | **0.2181**                 | **0.0054**                 | **0.1235**        | 0.768               | 0.0217                | No                 | 0.78             | No replay                      |
| **cvrp_random_replay**              | random                      | 0.2452                     | 0.0054                     | 0.1386            | 0.891               | 0.0061                | No                 | 0.58             | Random replay                  |
| **cvrp_failure_replay**             | failure                     | **0.2157 (best held-out**) | 0.0191                     | 0.1283            | 0.826               | **0.0572 (best)**     | No                 | 0.78             | Replay of failure cases        |
| **cvrp_failure_replay_compression** | failure + compression-aware | 0.2438                     | 0.0054                     | 0.1378            | **0.0 (lowest)*     | 0.0                   | Yes                | 0.58             | Replay + compression + novelty |

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### Conservative Interpretation

- **Held-out CVRPLIB Gap** (key metric for transfer):
  - Best is **cvrp\_failure\_replay** (0.2157), slightly better than no replay (0.2181).
  - Random replay and compression conditions perform worse (>0.24 gap).
- **Synthetic Holdout Gap** (transfer on synthetic data):
  - Lowest gaps (0.0054) for **no replay**, **random replay**, and **compression replay**.
  - Failure replay without compression shows noticeably worse synthetic gap (0.0191), indicating possibly less synthetic domain generality.
- **Mean Transfer Gap** (averaged transfer domain gap):
  - Lowest for **no replay** (0.1235), followed by failure replay (0.1283).
  - Random and compression-aware replay have higher transfer gaps (~0.138).

- **Code Novelty**:
  - Highest on random replay (0.891), followed by failure replay (0.826) and no replay (0.768).
  - Compression replay has zero novelty despite similar replay mode.
- **Adaptation Efficiency**:
  - Highest for failure replay (0.0572), lower for no replay (0.0217), and random replay (0.0061).
  - Compression-aware failure replay shows zero adaptation efficiency.
- **Replay mode and effects**:
  - Failure replay improves held-out CVRPLIB gap and adaptation efficiency compared to no replay.
  - Random replay leads to higher code novelty but worse optimality gaps.
  - Compression-aware replay reduces code novelty to zero and yields poor adaptation despite replaying failures.
- **Final Behavior Complexity**:
  - Higher complexity (0.78) for no replay and failure replay suggests more complex behavior profiles correlate with better gaps.
  - Lower complexity (0.58) for random and compression replay conditions aligned with worse generalization.

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## Important Points

- **Best transfer performance** (held-out CVRPLIB gap and synthetic gap) is with failure replay (no compression) and no replay conditions. No replay has the best synthetic and transfer gaps but slightly worse held-out CVRPLIB gap.
- **Code novelty decreases with compression-aware replay and fails to improve transfer despite replaying failure cases.**
- Random replay increases code novelty but results in worse gaps and lower adaptation efficiency.
- Replay mode substantially affects transfer ability: failure replay aids held-out generalization by improving optimality gap and adaptation efficiency, even though synthetic domain gaps are slightly worse.
- No replay condition shows a strong tradeoff with moderate code novelty and best synthetic transfer gaps.

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## Final Conservative Conclusions

- **Failure replay improves held-out CVRPLIB optimality gap and adaptation efficiency over no replay and random replay, indicating beneficial specialized replay of failure cases.**
- **No replay achieves the best synthetic holdout and lowest mean transfer gap, illustrating strong transfer despite lack of replay.**
- **Random replay increases code novelty but at the cost of worse transfer performance and adaptation efficiency.**

- **Compression-aware replay sharply decreases code novelty and adaptation efficiency, impairing transfer despite replaying failure cases.**
- **Overall, replaying failure cases without compression represents the best practical tradeoff between transfer optimality gap and adaptation.**
- **Lexical code novelty does not strictly correlate with transfer improvements, especially seen in compression failure replay condition where novelty drops but transfer is poor (no benefit in this case).**

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## **Recommendations for Benchmark Suite Use**

- Prioritize **failure replay without compression** for benchmarks aiming to optimize transfer on held-out CVRPLIB instances.
- Use **no replay** condition as a baseline for synthetic domain transfer quality.
- Treat **random replay** and **compression-aware replay** conditions cautiously, as increased novelty or replay compression do not guarantee transfer or optimality benefits.
- Emphasize deterministic metrics (held-out gaps, transfer gaps, adaptation efficiency) over code novelty for algorithmic assessment.